

February Crossword

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ONE ACROSS

On Valentine's Day, n labeled couples attend a dance, giving a total of $2n$ distinct people. They are to be arranged in a straight line for a group photograph. However, they are told that no couple may stand next to each other. Let a_n denote the number of such valid arrangements. Suppose the asymptotic behavior of a_n as $n \rightarrow \infty$ can be written in the formula $\frac{(2n!)}{c}$ for some constant c . Find the sum of the first four digits of c .

ONE DOWN

Romeo and Juliet communicate via a secure channel where their affection levels, represented by real numbers x and y , fluctuate according to a linked system of cubic equations. On Valentine's Day, their parameters satisfy:

$$x^3 - 3xy^2 = 46$$

$$y^3 - 3x^2y = 9$$

Let S be the magnitude of their combined bond, defined as $S = x^2 + y^2$. Find the value of S .

TWO DOWN

Twelve points are equally spaced on a circle of radius 10. Each point is connected to the point directly opposite it, forming six diameters. Adjacent diameters intersect at the center and create 12 congruent angles. Consider the region consisting of all points inside the circle whose distance to the center is less than their perpendicular distance to every diameter. If the area of this region is A , find $A + 87$.

THREE ACROSS

A computer dating algorithm assigns a "compatibility score" based on the sum of powers. Consider the integer S defined by the sum of the 2025-th powers of the first 2025 positive integers:

$$S = \sum_{k=1}^{2025} k^{2025} = 1^{2025} + 2^{2025} + \dots + 2025^{2025}.$$

If R is the remainder when S is divided by 2026, find the last two digits of R^{11} .